CSE416 — Software Project Design and Development (Lab)

SOFTWARE TEST REPORT

Test Plan · Test Specification · Test Completion Report

Project title	BloomVerse — Gardening Web Platform (shop, Plant Doctor, Swap Hub, Green Route delivery)
Team ID	09
Member 1 (ID, name, role)	0432320005101032 — Mitu Barua Omi — Developer
Member 2 (ID, name, role)	0432320005101050 — Zakia Sultana Chadni — Reviewer
Experiment	1
Instructor	Al Imtiaz, Associate Professor, Department of CSE
Date of submission	27/10/2026

Document Control

Repository URL	GitHub029-min/Software-Project-Design-and-Development
Version tested (tag / commit SHA)	Until it is tagged, the tested files can be identified by their SHA-256 (first 12 characters): api.php fd8249fa59a6, app.js 9f4583f1bdaf, database.sql 98b9f878b784, index.html d0d1afb6dbff, style.css 000a152cb036.
Test environment URL	All tests ran locally against a PHP server and a MariaDB test database.
Report version	1.0
Prepared by	Mitu Barua Omi & Zakia Sultana Chadni
Target tier (Guideline §3)	Baseline. Manual items that need a browser or other people (accessibility, compatibility, usability, DAST, CI link) are not done yet and are listed in section 11.

Revision history

Version	Date	Author	Changes
1.0	9/9/26	Mitu Barua Omi, Zakia Sultana Chadni	First complete report: automated unit and integration tests, coverage, load test, lint, security checks, 10 defects found and fixed.

1. Introduction

1.1 Purpose

This report explains how BloomVerse was tested, what was found, and whether the system is ready. It follows the course template and guideline. It is written for the course instructor and for the team, so a new person can run the same tests and get the same numbers.

1.2 Product overview

BloomVerse is a web application (HTML, CSS, JavaScript, PHP, MySQL) for plant lovers. Customers buy plants, seeds, fertilizer, flowers and tools with Cash on Delivery. The Plant Doctor checks a leaf photo and suggests a fix. The Swap Hub lets members give away or swap plants. The Green Route module lets a delivery rider go on shift, accept new orders and follow a planned route on a map. An admin manages products, riders, orders and reviews.

1.3 Scope of testing

In scope

Feature / module	Requirement IDs	Test levels applied
Authentication and sessions	FR-01, NFR-02, NFR-04	Integration (API + database), security checks
Shop: browse, search, filter	FR-02	Integration
Cart checkout, stock, Cash on Delivery	FR-03	Integration (transaction and boundary tests)
Plant Doctor (leaf analysis)	FR-04, NFR-03	Integration with generated images
Swap Hub	FR-05, NFR-03	Integration
Green Route: planner, shift, accept, deliver	FR-06, FR-07, FR-08	Unit (planner) and integration (rider flow)

Reviews and ratings	FR-10	Integration (full order to delivery journey)
Admin dashboard and product management	FR-09, FR-11	Integration
Performance, security, static analysis	NFR-01 to NFR-04	Load test, OWASP checklist, ESLint, php -l

Out of scope

Feature / item	Reason not tested
Map tiles (OpenStreetMap), road routing (OSRM), Leaflet and jsPDF internals	Third-party services and libraries. We only test our own use of them. The route display and PDF receipt need a browser and are listed as manual checks.
Screen layout and visual design in different browsers and phones	Needs a real browser. Planned for version 1.1 (section 8.4)
Online payment, live GPS tracking, mobile app	SRS section 1.3
Load above 20 users, HTTPS and server hardening	The project runs locally on XAMPP

1.4 References

Document	Version / link
SRS (Experiment 2)	BloomVerse SRS, updated version (requirement table in section 5.4, business rules in section 5.5). FR-10, FR-11 and NFR-04 were added in this report and should be added to the SRS table too.
Software Testing Report Guideline	CSE416_Testing_Report_Guideline.docx
Standards	ISO/IEC/IEEE 29119-3; ISO/IEC 25010; OWASP Top 10; WCAG 2.2

1.5 Definitions and abbreviations

FR / NFR: functional / non-functional requirement. TC: test case. EP / BVA: equivalence partitioning / boundary value analysis. P1: highest priority test. COD: Cash on Delivery. Depot: the fixed start point of every rider route. 2-opt: a method that shortens a route by reversing parts of it. PCOV: a PHP line-coverage tool. p95: 95 percent of requests were faster than this time.

2. Test Environment

2.1 Environment configuration

Operating system(s)	Ubuntu 24.04.4 LTS (container, 1 vCPU)
Runtime / language version	PHP 8.3.6 (CLI); Node.js 22.22.2
Framework(s)	No framework. Plain PHP with PDO; tests use Node.js built-in test runner (node:test) and fetch
Database (test instance)	MariaDB 10.11.14, database bloomverse_test. It is dropped and rebuilt from database.sql before every test file, so the real bloomverse database is never touched.
Browsers / devices	Microsoft Edge
Test hosting	PHP built-in server (php -S, 4 workers) on 127.0.0.1. Not XAMPP/Apache.
Test tools and versions	node:test (built in), PCOV 1.0.11 (coverage), ESLint 9.39.5, php -l, load script tests/performance/load.js, mutation script tests/mutation_spotcheck.py

2.2 Test data

- Product data: the 20 seeded products from database.sql.
- Accounts: admin@bloomverse.com (created by api.php), rider@bloomverse.com (seeded in database.sql.
- Images (leaf photos, swap photos) are generated by real photos
- Fixed random seeds are used for the planner tests, so results are repeatable.

2.3 How to run the tests

```
npm install          # installs ESLint only npm test          # unit + integration, about 15 s npm run test:coverage    # same + line coverage of api.php npm run test:perf        # 20 users for 30 s, prints p50/p95 npm run lint          # ESLint (also: php -l api.php)
```

Needs Node 22+, PHP 8 with gd and pdo_mysql (pcov for coverage) and a local MySQL/MariaDB with root and an empty password (XAMPP default). On Windows set PHP_BIN to the php.exe inside XAMPP if php is not on the PATH. The full steps are in TESTING_README.md.

3. Test Strategy

3.1 Test levels and types

Level / type	Objective	Technique(s)	Tool	Auto?	Owner
Unit	Check the route planner functions (distance and stop order)	EP, BVA, property tests, independent oracle, brute force	node:test	Yes	Team
Integration / API	Check every API action with a real PHP server and a real database	EP, BVA, decision table, state transition, error guessing	node:test + fetch + MariaDB	Yes	Team
System / E2E	Customer, rider and admin journeys through the browser	Use case	Manual (browser)	No	
Acceptance (UAT)	Check the SRS business rules with real users	Scenario	Manual	No	
Regression	Re-run all tests after every fix; each fixed defect has a test	All	node:test	Yes	Team

Level / type	Objective	Technique(s)	Tool	Auto?	Owner
Performance	Response time under 20 users	Load profile	tests/performance/load.js	Yes	Team
Security	Access control, injection, upload safety, errors, headers	OWASP checklist, error guessing	node:test + grep scan	Yes / partly	Team
Accessibility	Lighthouse and axe on key screens	Tool scan	Lighthouse	No	
Compatibility	Chrome, Firefox, phone size	Manual	Browsers	No	
Usability	Three users do three tasks	Task test	Manual	No	

3.2 Entry criteria

- The system starts and the database schema (database.sql) can be imported without errors.
- The SRS (with business rules in section 5.5) is available as the source of expected results.
- The test database can be rebuilt automatically, so every run starts from the same data.

3.3 Exit criteria

Criterion	Target	Achieved	Met?	Evidence
FRs with at least 1 test case	100%	11 of 11 (100%)	Yes	Section 4, A2
Priority-1 test cases passing	100%	20 of 20	Yes	Section 5.2, A2
Line coverage (core layer)	≥ 60%	api.php 99.3% (146/147 lines)	Yes	Section 7, A4
Branch coverage (core layer)	Not required at Baseline	Not measured (PCOV counts lines only)	n/a	Section 7
Open S1 / S2 defects	0 / 0	0 / 0	Yes	Section 9

Criterion	Target	Achieved	Met?	Evidence
Static-analysis quality gate	Linter: 0 errors	ESLint 0 problems; php -l clean	Yes	Section 8.3, A7
Dependency audit: Critical / High	0 / 0 unaddressed	No Critical or High confirmed. 1 Medium: jsPDF 2.5.1 is outdated (not reachable, see 8.2)	Yes (with note)	Section 8.2
p95 response time (key endpoints)	< 1000 ms (SRS 5.1)	Reads 716 to 747 ms; login 1094 ms	Partly	Section 8.1, A5
Lighthouse accessibility score	≥ 80 on key screens	Not run yet	No	Section 8.4
Other (SRS): usability with 3 users	Done	Not run yet	No	Section 8.5

4. Requirements Traceability Matrix

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
FR-01	Signup, login, logout	High	I01–I10	Integration	Passed
FR-02	Browse, search and filter products	High	I20–I26	Integration	Passed
FR-03	Cart and Cash on Delivery checkout	High	I30–I41	Integration	Passed (PDF receipt is a manual check)
FR-04	Plant Doctor leaf analysis	High	I50–I55, I69, I69b	Integration	Passed
FR-05	Swap Hub listings and interest	Medium	I60–I68	Integration	Passed
FR-06	Rider shift and accept order	High	I70–I77, I79	Integration	Passed

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
FR-07	Green Route planning (shortest route)	High	U01–U14, I83	Unit, Integration	Passed (map display is a manual check)
FR-08	Rider marks order delivered	High	I78, I80–I82	Integration	Passed
FR-09	Admin dashboard and moderation	Medium	I93–I96, I106	Integration	Passed
FR-10	Reviews and ratings (added to SRS)	Medium	I45–I49	Integration	Passed
FR-11	Admin adds, edits, removes products (added)	Medium	I97, I98, I100–I105	Integration	Passed
NFR-01	Fast page response (under 1 s per query, 3 s per page)	Medium	Load test (section 8.1)	Performance	Partly (login p95 1094 ms; page load not measured)
NFR-02	bcrypt passwords, prepared statements	High	I11–I13	Integration	Passed
NFR-03	Safe image upload	High	I56–I59, I99	Integration	Passed
NFR-04	Server-side role checks, session and error handling (SRS 5.3, added)	High	I14–I16, I90–I92, I107–I111	Integration	Passed
BR-1	Login needed to order, scan, post swaps	High	I33, I55, I65, I48	Integration	Passed
BR-2	Free adoption has no price	Low	By design: no price field exists	Review	n/a
BR-3	Only received products can be reviewed	High	I45, I46	Integration	Passed
BR-4	Only admin sets Eco label and makes riders	Medium	I90–I92, I94, I106	Integration	Passed

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
BR-5	Rider must be on shift to accept	High	I72	Integration	Passed
BR-6	One order belongs to one rider	High	I76, I79	Integration	Passed
BR-7	Rider sees only own orders	High	I77, I78	Integration	Passed
BR-8	Only owner or admin removes a swap listing	Medium	I67	Integration	Passed

Total requirements	Requirements with ≥1 test	Requirements with ≥1 negative test
15 (11 FR + 4 NFR) plus 8 business rules	15 of 15 (NFR-01 is covered only by the load test)	11 of 11 FR, and all 8 business rules except BR-2 (by design)

5. Test Case Specification

5.1 Test case summary

All test IDs are in the test names in the code (for example TC-I30). All cases are automated. Source "Tool" means the case was written with tool help (see section 10). U = unit, I = integration. TC-I34 and TC-I47 run several input values each.

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-U01	FR-07	Dist: same point is 0 km	Unit	BVA	P2	+	Yes	Tool	Pass
TC-U02	FR-07	Dist: 1 degree of longitude on the equator is about 111.19 km	Unit	Independent oracle	P2	+	Yes	Tool	Pass
TC-U03	FR-07	Dist: symmetric, A to B equals B to A	Unit	Property	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-U04	FR-07	Dist: agrees with independent haversine for Dhaka points	Unit	Independent oracle	P2	+	Yes	Tool	Pass
TC-U05	FR-07	Plan: no stops gives empty route of 0 km	Unit	BVA	P2	−	Yes	Tool	Pass
TC-U06	FR-07	Plan: one stop is the route, km equals depot to stop	Unit	BVA	P2	+	Yes	Tool	Pass
TC-U07	FR-07	Plan: stops on one line are visited near to far	Unit	Independent oracle	P1	+	Yes	Tool	Pass
TC-U08	FR-07	Plan: every stop appears exactly once (permutation property, 20 seeded sets)	Unit	Property	P2	+	Yes	Tool	Pass
TC-U09	FR-07	Plan: reported km equals the length of the returned order	Unit	Property	P2	+	Yes	Tool	Pass
TC-U10	FR-07	Plan: naive equals the length of the stops in the given order	Unit	Property	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-U11	FR-07	Plan: never longer than a plain nearest-stop route (2-opt only improves, 50 seeded sets)	Unit	Property	P1	+	Yes	Tool	Pass
TC-U12	FR-07	Plan: within 10% of the true best route for 6 stops (brute force, 10 seeded sets)	Unit	Brute-force oracle	P2	+	Yes	Tool	Pass
TC-U13	FR-07	Plan: does not change the array it was given	Unit	Error guessing	P2	+	Yes	Tool	Pass
TC-U14	FR-07	Plan: 30 stops finish in under 1 second (performance of the algorithm)	Unit	Performance	P3	+	Yes	Tool	Pass
TC-I01	FR-01	Register with valid data succeeds	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I02	FR-01	Register with an email that already exists is rejected	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I03	FR-01	Password of 5 characters is rejected	Integration	BVA	P2	−	Yes	Tool	Pass
TC-I04	FR-01	Password of exactly 6 characters is accepted	Integration	BVA	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I05	FR-01	Register without a name is rejected with a clear error	Integration	EP	P2	−	Yes	Tool	Pass
TC-I06	FR-01	Register with a malformed email is rejected	Integration	EP	P2	−	Yes	Tool	Pass
TC-I07	FR-01	Login with correct details returns id, name and role	Integration	Use case	P1	+	Yes	Tool	Pass
TC-I08	FR-01	Login with a wrong password is rejected	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I09	FR-01	Login with an unknown email is rejected with the same message	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I10	FR-01	Session: me returns the user after login and null after logout	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I11	NFR-02	Password is stored as a bcrypt hash, never as plain text	Integration	Checklist (OWASP)	P1	+	Yes	Tool	Pass
TC-I12	NFR-02	SQL injection in the login email does not log the attacker in	Integration	Error guessing (OWASP)	P1	−	Yes	Tool	Pass
TC-I13	NFR-02	SQL injection in product search is treated as plain text	Integration	Error guessing (OWASP)	P2	−	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I14	NFR-04	Stale session: a deleted account becomes anonymous on the next request	Integration	State transition	P2	+	Yes	Tool	Pass
TC-I15	NFR-04	Role change by an admin takes effect without logging in again	Integration	State transition	P2	+	Yes	Tool	Pass
TC-I16	NFR-04	Unknown action returns a JSON error	Integration	Use case	P3	+	Yes	Tool	Pass
TC-I20	FR-02	Product list returns all 20 seeded products with price, stock and category	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I21	FR-02	Category filter returns only that category (Plants = 4)	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I22	FR-02	Every section is browsable (Plants 4, Flowers 4, Seeds 4, Fertilizers 4, Tools 4)	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I23	FR-02	Search is case-insensitive and matches part of a name	Integration	Use case	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I24	FR-02	Search with no match returns an empty list	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I25	FR-02	Eco filter returns only Eco-Certified products (7 seeded)	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I26	FR-02	Category and search filters work together	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I30	FR-03	Order total is the sum of database prices times quantity, and stock goes down	Integration	Use case	P1	+	Yes	Tool	Pass
TC-I31	FR-03	A price sent by the browser is ignored, the database price is used	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I32	FR-03	Order status starts as Placed with no rider	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I33	FR-03	Ordering needs a login	Integration	Use case	P1	−	Yes	Tool	Pass
TC-I34	FR-03	Phone number format: 01XXXXXXXX with optional +88 (8 values, valid and invalid)	Integration	BVA	P2	±	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I35	FR-03	Address of 9 characters is rejected, 10 is accepted	Integration	BVA	P2	±	Yes	Tool	Pass
TC-I36	FR-03	Empty name is rejected	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I37	FR-03	Quantity above stock is rejected, stock is unchanged and no order row is kept	Integration	Use case	P1	−	Yes	Tool	Pass
TC-I38	FR-03	Quantity equal to stock is accepted (BVA: max), then one more is rejected	Integration	BVA	P2	±	Yes	Tool	Pass
TC-I39	FR-03	An order with no items is rejected	Integration	EP	P1	−	Yes	Tool	Pass
TC-I40	FR-03	An unknown product id is rejected and no order row is kept	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I41	FR-03	Myorders shows only the logged-in customer's own orders (access control)	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I45	FR-10	Review is rejected before delivery	Integration	State transition / use case	P2	−	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I46	FR-10	Review is accepted after the order is delivered, and shows with the average rating	Integration	State transition / use case	P1	+	Yes	Tool	Pass
TC-I47	FR-10	Rating is kept inside 1 to 5 (inputs 0, 1, 5, 6)	Integration	BVA	P2	±	Yes	Tool	Pass
TC-I48	FR-10	Anonymous user cannot review	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I49	FR-10	A flagged review disappears from the shop and appears in the admin queue; admin can delete it	Integration	State transition / use case	P2	±	Yes	Tool	Pass
TC-I50	FR-04	A healthy green leaf gives "Looks healthy" and a high health index	Integration	Decision table	P2	+	Yes	Tool	Pass
TC-I51	FR-04	A yellow leaf is reported as yellowing (watering/nutrient issue)	Integration	Decision table	P2	+	Yes	Tool	Pass
TC-I52	FR-04	A mostly brown leaf is reported as burn / dying tissue with a low index	Integration	Decision table	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I53	FR-04	Scan result gives care steps and a colour mix that adds up to about 100%	Integration	Use case	P3	+	Yes	Tool	Pass
TC-I54	FR-04	Every scan is saved in the history of that user only	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I55	FR-04	Plant Doctor needs a login	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I56	NFR-03	A GIF is rejected, only JPG and PNG are allowed	Integration	EP	P2	−	Yes	Tool	Pass
TC-I57	NFR-03	A text file renamed to .png is rejected	Integration	Error guessing	P2	−	Yes	Tool	Pass
TC-I58	NFR-03	A PHP script renamed to .jpg is rejected and never stored	Integration	Error guessing (OWASP)	P1	−	Yes	Tool	Pass
TC-I59	NFR-03	A file over 8 MB is rejected	Integration	BVA	P2	−	Yes	Tool	Pass
TC-I60	FR-05	A listing without a photo is created	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I61	FR-05	A listing with a photo is created and the photo can be fetched as JPEG	Integration	Use case	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I62	FR-05	A listing with an empty title is rejected	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I63	FR-05	An unknown type is saved as Free adoption (only two types allowed)	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I64	FR-05	A title of 150 characters is cut to 100	Integration	BVA	P2	+	Yes	Tool	Pass
TC-I65	FR-05	Creating a listing needs a login	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I66	FR-05	"Interested" counts each user once, even if pressed twice	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I67	FR-05	Only the owner or an admin can remove a listing	Integration	Use case	P2	±	Yes	Tool	Pass
TC-I68	FR-05	An image that does not exist returns 404	Integration	Use case	P3	+	Yes	Tool	Pass
TC-I69	FR-04	A leaf with green, yellow and dark patches is reported as mild leaf stress	Integration	Decision table	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I69b	FR-04	A leaf with green, yellow and brown/other spots is reported as leaf spot disease	Integration	Decision table	P2	+	Yes	Tool	Pass
TC-I70	FR-06	Customers cannot use rider actions (route, shift, accept, deliver)	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I71	FR-06	Anonymous users cannot use rider actions	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I72	FR-06	A rider starts off shift and cannot accept an order	Integration	State transition / use case	P2	−	Yes	Tool	Pass
TC-I73	FR-06	The new order is waiting in the list with area, address, total and location	Integration	State transition / use case	P2	+	Yes	Tool	Pass
TC-I74	FR-06	Shift can be switched off again	Integration	State transition / use case	P2	+	Yes	Tool	Pass
TC-I75	FR-06	Accepting moves the order to this rider with status "Out for delivery"	Integration	State transition / use case	P1	+	Yes	Tool	Pass
TC-I76	FR-06	A second rider cannot take an order that is already taken	Integration	State transition / use case	P1	−	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I77	FR-06	A rider sees route data only for his own orders	Integration	State transition / use case	P1	−	Yes	Tool	Pass
TC-I78	FR-08	A rider cannot mark another rider's order delivered	Integration	State transition / use case	P2	−	Yes	Tool	Pass
TC-I79	FR-06	Order accepted by two riders at the same moment goes to only one	Integration	State transition / use case	P2	+	Yes	Tool	Pass
TC-I80	FR-08	Delivering sets status Delivered, records the time, and updates today's statistics	Integration	State transition / use case	P1	+	Yes	Tool	Pass
TC-I81	FR-08	The customer sees the Delivered status in My orders	Integration	State transition / use case	P2	+	Yes	Tool	Pass
TC-I82	FR-08	A customer cannot jump the line: an order placed by another user stays unassigned until accepted	Integration	State transition / use case	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I83	FR-07	Route stops are returned in the same area as ordered and can be planned (3 orders accepted)	Integration	State transition / use case	P2	+	Yes	Tool	Pass
TC-I90	NFR-04	Customers get 403 on every admin action (broken access control)	Integration	Error guessing (OWASP)	P1	−	Yes	Tool	Pass
TC-I91	NFR-04	Riders get 403 on every admin action	Integration	Error guessing (OWASP)	P2	−	Yes	Tool	Pass
TC-I92	NFR-04	Anonymous users get 403 on every admin action	Integration	Error guessing (OWASP)	P1	−	Yes	Tool	Pass
TC-I93	FR-09	Dashboard statistics match the database	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I94	FR-09	Make rider: unknown email, normal user, and already-rider cases	Integration	Use case	P2	±	Yes	Tool	Pass
TC-I95	FR-09	Admin can assign an order to a rider; status becomes "Out for delivery"	Integration	Use case	P2	+	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I96	FR-09	Admin cannot assign an order to a user who is not a rider	Integration	Use case	P1	−	Yes	Tool	Pass
TC-I97	FR-11	Admin can add a product, and it shows in the shop	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I98	FR-11	Adding a product with an empty name is rejected	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I99	NFR-03	Adding a product with a non-image photo is rejected and nothing is added	Integration	Error guessing (OWASP)	P2	−	Yes	Tool	Pass
TC-I100	FR-11	A product with a negative price is rejected	Integration	EP	P1	−	Yes	Tool	Pass
TC-I101	FR-11	A product with negative stock is rejected	Integration	EP	P2	−	Yes	Tool	Pass
TC-I102	FR-11	Admin can edit name, category, price and stock of an existing (seeded) product	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I103	FR-11	Editing a product that does not exist returns an error	Integration	Use case	P2	−	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I104	FR-11	Removing a product hides it from the shop but keeps old orders readable	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I105	FR-11	A removed product cannot be ordered	Integration	Use case	P2	−	Yes	Tool	Pass
TC-I106	FR-09	Admin can switch the Eco-Certified label on and off	Integration	Use case	P2	+	Yes	Tool	Pass
TC-I107	NFR-04	Error responses do not reveal SQL or server details	Integration	Error guessing (OWASP)	P1	−	Yes	Tool	Pass
TC-I108	NFR-04	Responses to a normal request carry JSON content type	Integration	Use case	P3	+	Yes	Tool	Pass
TC-I109	NFR-04	An unexpected database error returns a generic message, not SQL text	Integration	Error guessing (OWASP)	P2	−	Yes	Tool	Pass
TC-I110	NFR-04	API responses carry basic security headers (X-Content-Type-Options, X-Frame-Options)	Integration	Checklist (OWASP)	P2	−	Yes	Tool	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-I111	NFR-04	The session cookie is HttpOnly and SameSite (cannot be read by page scripts)	Integration	Checklist (OWASP)	P2	–	Yes	Tool	Pass

5.2 Detailed test cases (all priority-1 cases)

Test case ID / title	TC-I07 — Login with correct details returns id, name and role
Requirement / acceptance criterion	FR-01: login returns the user; SRS 2.2 (1)
Level / technique / priority	Integration / use case / P1
Preconditions	Account ayesha@test.example (role user) exists
Test data	email ayesha@test.example, password secret1
Steps	1. POST login with the correct email and password 2. Read the response
Expected result (from SRS)	Response has id, name "Ayesha", role "user"
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/auth.test.js › "login with correct details returns id, name and role"
Executed by / date	Mitu, 20-10-2026

Test case ID / title	TC-I11 — Password is stored as a bcrypt hash, never as plain text
Requirement / acceptance criterion	NFR-02: passwords hashed with bcrypt (SRS 5.3)
Level / technique / priority	Integration / checklist / P1

Preconditions	Account ayesha@test.example registered
Test data	password secret1
Steps	1. Register the account 2. Read the password column directly from the database
Expected result (from SRS)	Stored value starts with $2y$ and is not equal to secret1
Actual result	Matches expected. Passed on every run.
Status	Pass
Automated test location	tests/integration/auth.test.js › "password is stored as a bcrypt hash"
Executed by / date	Omi, 20-10-2026

Test case ID / title	TC-I12 — SQL injection in the login email does not log the attacker in
Requirement / acceptance criterion	NFR-02: SQL injection protection (SRS 5.3)
Level / technique / priority	Integration / error guessing / P1
Preconditions	Server running
Test data	email = ' OR 1=1 -- , password = x
Steps	1. POST login with the injection text as email 2. Check the response and session
Expected result (from SRS)	Login is rejected; no user id is returned
Actual result	Matches expected. Passed on every run.
Status	Pass
Automated test location	tests/integration/auth.test.js › "SQL injection in the login email"
Executed by / date	Zakia, 22-10-2026

Test case ID / title	TC-I30 — Order total is the sum of database prices times quantity, and stock goes down
Requirement / acceptance criterion	FR-03: order total and stock (SRS 2.2 (6))
Level / technique / priority	Integration / use case / P1
Preconditions	Logged-in customer; Trowel (350, stock 15), Gloves (250, stock 25)
Test data	Trowel x2, Gloves x1, valid delivery details
Steps	1. Place the order 2. Read total and both stock values
Expected result (from SRS)	ok = 1; total = 350×2 + 250 = 950; stock drops by 2 and 1
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/shop.test.js › "order total is the sum of database prices times quantity"
Executed by / date	Zakia, 23-10-2026

Test case ID / title	TC-I33 — Ordering needs a login
Requirement / acceptance criterion	BR-1: login required to order (SRS 5.5)
Level / technique / priority	Integration / use case / P1
Preconditions	No login
Test data	Trowel x1
Steps	1. POST order without a session
Expected result (from SRS)	HTTP 403 and no order is created

Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/shop.test.js › "ordering needs a login"
Executed by / date	Omi, 24-10-16

Test case ID / title	TC-I37 — Quantity above stock is rejected, stock is unchanged and no order row is kept
Requirement / acceptance criterion	FR-03: stock check inside a transaction
Level / technique / priority	Integration / BVA / P1
Preconditions	Customer logged in; Sunflower stock = S
Test data	quantity = S + 1
Steps	1. Order Sunflower with quantity S+1 2. Count orders and read stock
Expected result (from SRS)	Error "Not enough stock"; stock unchanged; order count unchanged (rolled back)
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/shop.test.js › "quantity above stock is rejected"
Executed by / date	24-10-16

Test case ID / title	TC-I39 — An order with no items is rejected
Requirement / acceptance criterion	FR-03: an order must contain items

Level / technique / priority	Integration / EP / P1
Preconditions	Customer logged in
Test data	items = [] (empty cart)
Steps	1. POST order with an empty items list 2. Count orders
Expected result (from SRS)	Error "Your cart is empty"; no order row is created
Actual result	Matches expected. First run FAILED (defect found, see section 9); passes after the fix.
Status	Pass
Automated test location	tests/integration/shop.test.js › "an order with no items is rejected"
Executed by / date	

Test case ID / title	TC-I46 — Review is accepted after the order is delivered, and shows with the average rating
Requirement / acceptance criterion	BR-3: only received products can be reviewed
Level / technique / priority	Integration / state transition / P1
Preconditions	Order placed, accepted and delivered by the rider
Test data	rating 4, comment "Healthy plant"
Steps	1. Customer orders Lily 2. Rider goes on shift, accepts, delivers 3. Customer posts the review
Expected result (from SRS)	Review accepted; shown in the review list; average rating 4
Actual result	Matches expected. Passed on every run.

Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/shop.test.js › "review is accepted after the order is delivered"

Test case ID / title	TC-I58 — A PHP script renamed to .jpg is rejected and never stored
Requirement / acceptance criterion	NFR-03: uploads are checked (SRS 5.2)
Level / technique / priority	Integration / error guessing / P1
Preconditions	Customer logged in
Test data	File shell.jpg containing "<?php echo 1; ?>"
Steps	1. Upload the file to the Plant Doctor
Expected result (from SRS)	Error "Upload a JPG or PNG photo"; file is not stored or run
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/doctor_swap.test.js › "a PHP script renamed to .jpg is rejected"

Test case ID / title	TC-I75 — Accepting moves the order to this rider with status "Out for delivery"
Requirement / acceptance criterion	FR-06: accept moves the order to the rider (SRS 4.1)
Level / technique / priority	Integration / state transition / P1
Preconditions	Rider on shift; order Placed
Test data	order id

Steps	1. Rider calls accept 2. Rider reads route
Expected result (from SRS)	Order is in rider orders with status "Out for delivery" and gone from the waiting list; phone number visible
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/rider.test.js › "accepting moves the order to this rider"

Test case ID / title	TC-I76 — A second rider cannot take an order that is already taken
Requirement / acceptance criterion	BR-6: one order, one rider
Level / technique / priority	Integration / state transition / P1
Preconditions	Order already accepted by rider 1; rider 2 on shift
Test data	order id
Steps	1. Rider 2 calls accept for the same order
Expected result (from SRS)	Error "already taken"; order still belongs to rider 1
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/rider.test.js › "a second rider cannot take an order that is already taken"

Test case ID / title	TC-I77 — A rider sees route data only for his own orders
Requirement / acceptance criterion	BR-7: rider sees only own orders (SRS 5.5)

Level / technique / priority	Integration / use case / P1
Preconditions	Order belongs to rider 1
Test data	rider 2 session
Steps	1. Rider 2 calls route
Expected result (from SRS)	No orders of rider 1 in the list
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/rider.test.js › "a rider sees route data only for his own orders"

Test case ID / title	TC-I80 — Delivering sets status Delivered, records the time, and updates today's statistics
Requirement / acceptance criterion	FR-08: mark delivered (SRS 4.1)
Level / technique / priority	Integration / state transition / P1
Preconditions	Order accepted by the rider
Test data	order id (total 1000)
Steps	1. Rider calls deliver 2. Read the order in the database and the rider statistics
Expected result (from SRS)	Status Delivered; delivered_at is set; delivered today +1; cash +1000; order leaves the route
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/rider.test.js › "delivering sets status Delivered, records the time"

Test case ID / title	TC-I90 — Customers get 403 on every admin action (broken access control)
Requirement / acceptance criterion	NFR-04: access control (SRS 2.3.4)
Level / technique / priority	Integration / error guessing / P1
Preconditions	Customer logged in
Test data	All 9 admin actions
Steps	1. Customer calls each admin action
Expected result (from SRS)	HTTP 403 for every action
Actual result	Matches expected. Passed on every run.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/admin_security.test.js › "customers get 403 on every admin action"
Executed by / date	Omi

Test case ID / title	TC-I92 — Anonymous users get 403 on every admin action
Requirement / acceptance criterion	NFR-04: access control
Level / technique / priority	Integration / error guessing / P1
Preconditions	No login
Test data	All 9 admin actions
Steps	1. Call each admin action without a session
Expected result (from SRS)	HTTP 403 for every action
Actual result	Matches expected. Passed on every run.

Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/admin_security.test.js › "anonymous users get 403 on every admin action"

Test case ID / title	TC-I96 — Admin cannot assign an order to a user who is not a rider
Requirement / acceptance criterion	FR-09: only riders can be assigned (BR-4)
Level / technique / priority	Integration / EP / P1
Preconditions	Placed order; a normal customer account
Test data	rider = id of the customer
Steps	1. Admin calls assign with a customer id
Expected result (from SRS)	Error; order keeps no rider
Actual result	Matches expected. First run FAILED (defect found, see section 9); passes after the fix.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/admin_security.test.js › "admin cannot assign an order to a user who is not a rider"

Test case ID / title	TC-I100 — A product with a negative price is rejected
Requirement / acceptance criterion	FR-11: price must be 0 or more
Level / technique / priority	Integration / EP / P1
Preconditions	Admin logged in
Test data	name "Neg price", price -5, stock 1
Steps	1. Admin adds the product

Expected result (from SRS)	Error; product is not created
Actual result	Matches expected. First run FAILED (defect found, see section 9); passes after the fix.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/admin_security.test.js › "a product with a negative price is rejected"

Test case ID / title	TC-I107 — Error responses do not reveal SQL or server details
Requirement / acceptance criterion	NFR-04: errors must not show SQL (SRS 5.3)
Level / technique / priority	Integration / error guessing / P1
Preconditions	Admin logged in
Test data	category "NoSuchCategory", price "abc"
Steps	1. Add a product with invalid values 2. Read the full response text
Expected result (from SRS)	Clean error; no SQLSTATE, PDOException or file name in the response
Actual result	Matches expected. First run FAILED (defect found, see section 9); passes after the fix.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/integration/admin_security.test.js › "error responses do not reveal SQL or server details"

Test case ID / title	TC-U07 — Plan: stops on one line are visited near to far
Requirement / acceptance criterion	FR-07: shortest route (SRS 4.1 algorithm)

Level / technique / priority	Unit / independent oracle / P1
Preconditions	DEPOT fixed in app.js
Test data	Three stops on one line north of the depot: far, near, mid
Steps	1. Call plan([far, near, mid])
Expected result (from SRS)	Order is near, mid, far; route length equals the straight distance to the farthest stop (within 0.01 km)
Actual result	Matches expected. Passed on every run.
Status	Pass
Automated test location	tests/unit/route.test.js › "stops on one line are visited near to far"
Executed by / date	Zakia

Test case ID / title	TC-U11 — Plan: never longer than a plain nearest-stop route (2-opt only improves, 50 seeded sets)
Requirement / acceptance criterion	FR-07: 2-opt only improves the route
Level / technique / priority	Unit / property / P1
Preconditions	50 seeded random stop sets (4 to 10 stops)
Test data	Fixed seeds 100 to 149
Steps	1. Compute plan() length 2. Compute a plain nearest-stop route with separate code
Expected result (from SRS)	plan() is never longer than the nearest-stop route
Actual result	Matches expected. Passed on every run.
Status	Pass
Automated test location	tests/unit/route.test.js › "never longer than a plain nearest-stop route"

Executed by / date	Omi

5.3 Equivalence partitions and boundary values

Input	Valid partition(s)	Invalid partition(s)	Boundary values tested	TC IDs
Password length	6 or more characters	Fewer than 6	5 (rejected), 6 (accepted)	I03, I04
Phone number	01 + 9 digits, optional +88, spaces/dashes ignored	Wrong length, wrong prefix	10 digits, 11 digits, 12 digits; +880…; 880…; 1690…; 02…	I34
Delivery address length	10 or more characters	Fewer than 10	9 (rejected), 10 (accepted)	I35
Order quantity	1 up to stock	More than stock, empty cart, unknown product	stock (accepted), stock + 1 (rejected), 0 items	I37, I38, I39, I40
Rating	1 to 5	Below 1, above 5 (stored as 1 or 5)	0, 1, 5, 6	I47
Leaf photo file	JPG or PNG up to 8 MB	GIF, text file, PHP file, over 8 MB	GIF, fake .png, fake .jpg, 8.1 MB	I56–I59
Swap title length	1 to 100 characters	Empty, over 100 (cut to 100)	Empty, 150 characters	I62, I64
Product price and stock (admin)	0 or more, numbers	Negative, text	price -5, stock -3, price "abc"	I100, I101, I107
Stops on a route	1 or more	0 stops	0, 1, 2 to 10, 30 stops	U05, U06, U08, U14

5.4 Decision table / state transition

Plant Doctor decision table (rules are in api.php; the SRS gives no numbers)

Rule (checked in this order)	Condition on the colours counted in the leaf	Result	TC ID
1	Brown/other + dark share is 30% or more	Leaf burn or dying tissue	I52
2	Brown/other ≥ 5% and (yellow ≥ 12% or dark ≥ 2%)	Leaf spot disease	I69b
3	Yellow ≥ 35%	Yellowing leaves	I51
4	Health index ≥ 70	Looks healthy	I50
5	None of the above	Mild leaf stress	I69

Order status transitions

From	Event	To	TC ID
Placed	Rider on shift accepts (or admin assigns)	Out for delivery	I75, I95
Placed	Rider off shift tries to accept	Stays Placed (error)	I72
Out for delivery	Another rider tries to accept / deliver	No change	I76, I78
Out for delivery	Owner rider delivers	Delivered	I80
Placed / Out for delivery	Customer tries to review	Review rejected	I45
Delivered	Customer reviews	Review saved	I46

6. Test Execution Results

6.1 Summary

Level / type	Total	Passed	Failed	Blocked	Skipped	Pass %
Unit	14	14	0	0	0	100%
Integration / API	107 (97 test cases; two cases run several inputs)	107	0	0	0	100%
System / E2E	Not run	-	-	-	-	-

Level / type	Total	Passed	Failed	Blocked	Skipped	Pass %
Acceptance	Not run	-	-	-	-	-
Total	121	121	0	0	0	100%

Final run: 121 tests, 121 passed, 0 failed, 0 skipped, 13 seconds. Four full runs in a row were green (flakiness check). The very first run, before any fix, had 116 tests with 109 passed and 7 failed. These 7 failures were real defects (section 9).

6.2 Continuous-integration evidence

Workflow file	.github/workflows/test.yml
CI run URL (tagged commit)	GitHub029-min/Software-Project-Design-and-Development
Commit SHA of run	SHA
Run status	Not run yet. The same commands passed locally (evidence A2, A3).
Branch protection requires checks	Yes
Consecutive green runs (flakiness check)	4 local runs, all 121 of 121 green (evidence A3).

6.3 Analysis of failed, blocked or skipped tests

Nothing fails, is blocked or skipped in the final run. These tests failed first and were fixed:

TC ID	Status	Reason	Linked defect	Action / plan
TC-I39	Failed, now pass	Empty cart created a Tk 0 order	D-01	Fixed in api.php
TC-I05	Failed, now pass	Missing name gave "Email already used"	D-02	Fixed
TC-I06	Failed, now pass	Malformed email accepted	D-03	Fixed
TC-I96	Failed, now pass	Order assigned to a non-rider account	D-04	Fixed

TC ID	Status	Reason	Linked defect	Action / plan
TC-I100	Failed, now pass	Negative price accepted	D-05	Fixed
TC-I101	Failed, now pass	Negative stock accepted	D-06	Fixed
TC-I107, I109	Failed, now pass	SQL error text shown to the user	D-07	Fixed
TC-I110	Failed, now pass	No security headers	D-08	Fixed
TC-I111	Failed, now pass	Session cookie not HttpOnly / SameSite	D-09	Fixed

7. Code Coverage

Module / layer	Line %	Branch %	Function %	Target	Met?
Business logic + controllers + data access (api.php: one file with inline SQL)	99.3% (146 of 147 lines)	Not measured	Not measured	≥ 60% line	Yes
Route helpers in app.js (dist, plan)	Not measured; all paths are exercised by 14 unit tests	Not measured	Not measured	-	n/a
UI components (rest of app.js)	Not measured; no browser tests yet	Not measured	Not measured	-	No
Overall	Not computed	-	-	-	-

Uncovered code — justification

One line is not covered. It is line 153, which is only the closing end of the file and holds no code, so no executable line is missed. In the first coverage run the "Mild leaf stress" branch of the Plant Doctor was missed. Tests TC-I69 and TC-I69b were added to cover it. PCOV counts lines, not branches. api.php has many statements on one line, so line coverage looks higher than branch coverage would

be. The real coverage of the front end (app.js) is low because it has no browser tests. Source: evidence A4.

Mutation testing (Distinction tier)

Tool / module	Mutants generated / killed / survived	Mutation score	Survivors analysed (examples)
Manual spot check (tests/mutation_spotcheck.py): 10 changes in api.php, 2 in app.js	12 / 11 / 1	92% (not a full mutation tool)	M06: removing "rider_id IS NULL" from the accept query survives. This is an equivalent mutant: every order that has a rider already has status "Out for delivery", and accept also requires status "Placed". It cannot change behaviour. The 11 others were caught (for example password limit 6→5 by TC-I03, stock check by TC-I38, inverted owner check by TC-I67).

This is a spot check, not a full mutation score. Distinction tier is not claimed.

8. Non-Functional Testing

8.1 Performance

Scenario	Load profile	Target (NFR)	p95 measured	Error rate	Result
GET products (shop page data)	20 users, 30 s, mixed scenarios	< 1000 ms (SRS 5.1)	746 ms (p50 256 ms)	0%	Met
GET products with search and category	same	< 1000 ms	716 ms (p50 254 ms)	0%	Met
GET swaps (Swap Hub)	same	< 1000 ms	747 ms (p50 221 ms)	0%	Met

Scenario	Load profile	Target (NFR)	p95 measured	Error rate	Result
POST login (bcrypt)	same	< 1000 ms	1094 ms (p50 658 ms)	0%	Not met

Total: about 1,500 requests, 50.9 requests per second, 0 errors (evidence A5). The test ran on 1 CPU that was shared by the PHP server, MariaDB and the load script, so the numbers are worse than a normal computer. Login is slowest because bcrypt is made slow on purpose. Please repeat this test on your own machine. Page load time in a browser (the 3 second target in NFR-01) was not measured. Load test results are not exact, but they show that the API itself is fast.

A load test also helped find D-10: every request ran 4 setup queries (checking table columns and the admin account). They now run once per session. Throughput went from 45.6 to 50.9 requests per second.

8.2 Security testing (completed in Experiment 10)

Check	Tool	Findings C / H / M / L	Action taken	Status
Dependency vulnerability audit	Manual review (no npm packages in the product; browser libraries: jsPDF 2.5.1, Leaflet 1.9.4 from CDN)	0 / 0 / 1 / 0	jsPDF 2.5.1 is old and has published advisories. The advisories are about features BloomVerse does not use (addImage, html); app.js has no call to them. Plan: update to the current jsPDF release. Not updated now, so the receipt PDF can be tested first.	Open (Medium, documented)

Check	Tool	Findings C / H / M / L	Action taken	Status
Secret scanning	grep patterns (gitleaks not available here)	0 / 0 / 2 / 0	No API keys or tokens found. Found default passwords in the code: admin123 (api.php), rider123 (database.sql), and database user root with an empty password. Fine for local use; must be changed before any real deployment.	Accepted for local use
Dynamic scan (DAST)	OWASP ZAP	Not run	ZAP could not be run in the test environment	Pending
Other	Static scan for unescaped user text in HTML (app.js)	0 / 0 / 0 / 0	24 places write HTML; all user fields go through esc(). The one candidate on line 32 is an image name reduced to letters and digits by a regex.	Done

OWASP-based manual security test checklist

Risk area	Test performed	TC ID	Result
Broken access control	Customer, rider and anonymous calls to admin, rider and customer-only actions; reading other users' orders and scans; removing other users' swap listings	I41, I54, I67, I70, I71, I77, I78, I90–I92	Pass
Authentication failures	Wrong password, unknown email, stale session after deleting a user, role change; cookie flags. Note: password minimum is only 6 characters and there is no lock after many wrong tries.	I08, I09, I14, I15, I111	Pass (2 observations)
Injection	SQL injection text in login email and in product search	I12, I13	Pass

Risk area	Test performed	TC ID	Result
Cross-site scripting (XSS)	Static scan of app.js (done). Browser test with a name like <img onerror> is not done yet.	- (evidence A7)	Partly done
Cryptographic failures	Password is stored as bcrypt, not plain text. HTTPS is not used on the local site.	I11	Pass (HTTPS not applicable locally)
Security misconfiguration	SQL text in error replies, missing security headers, uploaded PHP file, default passwords	I58, I99, I107, I109, I110	Pass after fixes; default passwords still open
Vulnerable components	Review of jsPDF and Leaflet versions	-	Open (Medium)

8.3 Static analysis and quality gate (completed in Experiment 10)

Metric	Value	Gate threshold	Pass?
Bugs	ESLint: 0 errors, 0 warnings (3 unused-variable errors in app.js were fixed). php -l: no syntax errors	0	Yes
Vulnerabilities	0 found by static scan; SonarQube was not run	0	Yes (limited tool)
Security hotspots reviewed	Default passwords and DB root user reviewed (section 8.2)	All reviewed	Yes
Code smells	Not measured (no SonarQube)	-	n/a
Duplication %	Not measured	≤ 3%	n/a
Coverage on new code	api.php 99.3%	≥ 60%	Yes
Reliability / Security / Maintainability rating	Not measured	A / A / A	n/a
Quality gate status	Passed on the checks that were run (ESLint, php -l)	Passed	Yes (limited)

9. Defect Report

9.1 Summary by severity

Severity	Found	Fixed & verified	Open	Deferred (justified)
S1 — Critical	0	0	0	0
S2 — Major	4	4	0	0
S3 — Minor	5	5	0	0
S4 — Trivial	1	1	0	0
Total	10	10	0	0

Defects found by own tests	Defects found by cross-team testing (label found-by:)	Defects found by tools (scanners / generators)
10	0	0

9.2 Defect log

Issue #	Title	Sev	Pri	Found by (TC / level)	Status	Regression TC
D-01	Order with an empty cart is accepted (creates a Tk 0 order that riders can see)	S2	P1	TC-I39 / Integration	Fixed	TC-I39
D-02	Register without a name says "Email already used"	S3	P2	TC-I05	Fixed	TC-I05
D-03	Register accepts an email like "not-an-email"	S3	P2	TC-I06	Fixed	TC-I06
D-04	Admin can assign an order to a customer account or a missing order	S2	P1	TC-I96	Fixed	TC-I96
D-05	Admin can create a product with a negative price	S2	P1	TC-I100	Fixed	TC-I100

Issue #	Title	Sev	Pri	Found by (TC / level)	Status	Regression TC
D-06	Admin can create a product with negative stock	S3	P2	TC-I101	Fixed	TC-I101
D-07	Database error text (SQLSTATE, table names) is sent to the browser	S2	P1	TC-I107, TC-I109	Fixed	TC-I107, TC-I109
D-08	No security headers on API replies	S3	P2	TC-I110	Fixed	TC-I110
D-09	Session cookie is not HttpOnly and has no SameSite	S3	P2	TC-I111	Fixed	TC-I111
D-10	Every request runs 4 database setup queries (slower under load)	S4	P3	Load test (performance)	Fixed	Load test re-run (50.9 vs 45.6 req/s)

9.3 Detailed reports for S1 and S2 defects

Issue # / title	D-01 — Order with an empty cart is accepted
Steps to reproduce	Log in as a customer. POST the order action with valid delivery details and items = [].
Expected	The order is rejected with a clear message (an order needs at least one item).
Actual	ok = 1 and an order with total 0 is created with status Placed. Riders see it in "New orders waiting".
Environment / version	PHP 8.3.6, MariaDB 10.11.14, files as listed in Document Control (before fix)
Severity justification	S2: wrong result and it disturbs riders, but it does not lose money or data.
Root cause	The code only checked each item inside a loop; with no items the loop never ran.
Fix (link)	api.php now returns "Your cart is empty" before the transaction starts.

Regression test	TC-I39 failed before, passes after

Issue # / title	D-04 — Order can be assigned to a user who is not a rider
Steps to reproduce	Admin calls a_assign with rider = id of a normal customer.
Expected	Only an account with role rider can be assigned; a missing or delivered order is rejected.
Actual	The order got rider_id of the customer and status "Out for delivery". It left the waiting list and no rider could deliver it.
Environment / version	PHP 8.3.6, MariaDB 10.11.14, files as listed in Document Control (before fix)
Severity justification	S2: a customer order can get stuck forever.
Root cause	a_assign ran the UPDATE without checking the rider id or the order.
Fix (link)	api.php checks the rider role and that the order exists and is not delivered.
Regression test	TC-I96 failed before, passes after

Issue # / title	D-05 — Product with a negative price
Steps to reproduce	Admin adds a product with price -5.
Expected	Price must be a number of 0 or more.
Actual	The product was saved with a negative price. An order for it would give a negative total.
Environment / version	PHP 8.3.6, MariaDB 10.11.14, files as listed in Document Control (before fix)
Severity justification	S2: wrong money values; the SRS expects correct totals. Only an admin can do it, so it is not S1.
Root cause	No server-side validation of price or stock.
Fix (link)	api.php validates name, category, price and stock for add and edit (new function pvalid).
Regression test	TC-I100 and TC-I101 failed before, pass after

Issue # / title	D-07 — SQL error text shown to users
Steps to reproduce	Admin adds a product with price "abc" and a wrong category (or place an order with an area of 100 characters).
Expected	A short message that does not show database or file details (SRS 5.3 security).
Actual	The reply contained the raw database error (SQLSTATE and column details).
Environment / version	PHP 8.3.6, MariaDB 10.11.14, files as listed in Document Control (before fix)
Severity justification	S2: security weakness (information disclosure). It does not give direct access, so it is not S1.
Root cause	The global exception handler printed the exception message.
Fix (link)	The handler now writes the detail to the PHP error log and returns "Server error. Please try again."
Regression test	TC-I107 and TC-I109 failed before, pass after

10. Tool-Assisted Testing Log

#	Tool	Asked / generated	Own version first?	How the expected value was verified	Decision	Reason
1	Claude (Anthropic)	The 14 unit tests for dist() and plan(), including the helper that loads the functions from app.js	No	Expected values from geometry (1° on the equator is about 111.19 km), an independent haversine, and a brute-force best route written separately	Accepted after running	All passed; a mutation that picked the farthest stop was caught (TC-U11)

#	Tool	Asked / generated	Own version first?	How the expected value was verified	Decision	Reason
2	Claude (Anthropic)	The 107 API integration tests and helpers (server start, database reset, image generator)	No	Expected values from the SRS business rules and field limits in the code (for example phone format, 6-character password); each test was also checked with planted bugs	Edited	Two tests were wrong at first: the Plant Doctor test images were blurred by image scaling, so the colour shares were wrong. The stripes were made wider.
3	Claude (Anthropic)	Found and fixed the 10 defects in api.php and app.js (3 unused variables)	No	Each failing test was shown failing on the old code, then passing on the new code	Accepted	See section 9
4	Claude (Anthropic)	Load test, coverage and mutation scripts; lint and security scans; this report draft	No	Numbers copied from the real run output files in evidence A2 to A7	Edited	

Reflection

Most of the tests in this project were generated by a tool, so the team must be able to explain every test case. These points are true for this work:

- One generated test was wrong. The first version of the Plant Doctor tests used narrow colour stripes. The image is shrunk to 80 × 80 pixels before analysis, so the stripes blurred and the result was wrong. We found it because the test failed, and we changed the test, not the code.
- Expected values for the Plant Doctor come from the thresholds in the code, because the SRS gives no numbers. This is a known weak point. Everything else uses the SRS business rules or independent calculations.

- One planted bug survived the spot check (M06). We analysed it and found it is an equivalent change, so the tests are not missing anything there.
- The tests found 10 real defects, so the suite is not only confirming what the code already does.

11. Test Completion Summary

11.1 Exit-criteria evaluation

Target tier is Baseline. The Baseline criteria for functional testing are met: all 11 FRs have tests, all 20 P1 tests pass, api.php line coverage is 99.3%, there are no open S1 or S2 defects, and the linter has 0 errors. These parts are not met yet and need to be done by the team: Lighthouse accessibility (target 80), at least two browsers, a usability test with 3 people, a CI run link, and the login response time (p95 1094 ms on a 1-CPU machine; target under 1 s). Details are in table 3.3.

11.2 Residual risks and known issues

Issue / risk	Impact on users	Likelihood	Mitigation / workaround
Default passwords (admin123, rider123) and database user root with no password	Anyone could take over the admin account if the site is put online	High if deployed; low on a local computer	Change both passwords and create a database user before any deployment
jsPDF 2.5.1 is outdated	Possible problem in PDF generation. BloomVerse does not call the affected features	Low	Update to the current release and test the receipt
No browser tests, no accessibility, compatibility or usability results	Layout or accessibility problems may exist that API tests cannot see	Medium	Do section 8.4 and 8.5 and the manual cases below
Plant Doctor rules are simple colour rules, not machine learning	A leaf photo with unusual lighting can give a wrong result	Medium	The SRS describes it as rule-based. Show a note in the app that it is only advice

Issue / risk	Impact on users	Likelihood	Mitigation / workaround
Login is slow under 20 users on a 1-CPU machine; no limit on wrong password tries	Slow login at peak time; guessing passwords is possible	Low to medium	Repeat the load test on the project machine; add a limit later
Manual checks not run: PDF receipt download, map display and route line, flip-card login screen	A UI bug could hide behind passing API tests	Medium	Run the three manual cases once in Chrome and record screenshots

11.3 Release recommendation

Conditional go for the course demonstration on a local computer: the backend behaves as the SRS says, every found defect is fixed and has a regression test. **No go** for a public release until the default passwords are changed, jsPDF is updated and the manual browser checks are done.

11.4 Lessons learned

- Writing the boundary tests (phone, address, stock, rating) found most of the defects early.
- A test that fails on the old code and passes on the new code is the best proof that a fix works. We did this for every defect.
- A load test found a small design problem (setup queries on every request) that normal tests would not show.
- Next time: write the test list from the SRS before the code is finished, and test in the browser at the same time as the API.

12. Individual Contribution

Member	Test cases designed (IDs)	Tests automated	Defects found	Defects fixed	Test-related PRs
Mitu Barua Omi (0432320005101032)					
Zakia Sultana Chadni (0432320005101050)					

13. Sign-off and Declaration

We declare that the results in this report come from the tagged version, that tool-assisted work is listed in section 10, and that we can explain every test case.

	Name	Signature	Date
Member 1	Mitu Barua Omi	Mitu	27-10-2026
Member 2	Zakia Sultana Chadni	zakia	27-10-2026

Appendix A — Evidence Index

#	Evidence	File / URL	Supports section
A1	First full run before fixes (116 tests, 7 failed) and defect before/after list	docs/testing/evidence/run_before_fixes.tap, defects_before_after.txt	6.3, 9
A2	Final full run (121 of 121 pass)	docs/testing/evidence/run_final.tap	3.3, 5, 6.1
A3	Three more full runs (flakiness check)	docs/testing/evidence/flakiness_run1.tap to run3.tap	6.2
A4	Line coverage of api.php (PCOV)	docs/testing/evidence/coverage_api_php.txt	7
A5	Load test output	docs/testing/evidence/performance_load.txt	8.1
A6	Mutation spot check output	docs/testing/evidence/mutation_spotcheck.txt	7
A7	ESLint, php -l, secret scan, XSS scan	docs/testing/evidence/lint_and_static.txt	8.2, 8.3
A8	Test source code and CI workflow	tests/, .github/workflows/test.yml, TESTING_README.md	2.3, 5